I&EC research
Industrial & Engineering Chemistry Research

Article

pubs.acs.org/IECR

# Design and Operation Strategy of $CO_2$ Terminal

Umer Zahid,[†] Jinjoo An,[†] Chul-Jin Lee,[‡] Ung Lee,[†] and Chonghun Han*,[†]

[†]School of Chemical and Biological Engineering, Seoul National University, Seoul 151-744, Republic of Korea

[‡]Seoul National University Engineering Research Institute, Seoul 151-744, Republic of Korea

**ABSTRACT:** With recent advancements in the carbon capture and storage (CCS), much interest has been developed in the $CO_2$ transport options. An infrastructure including pipelines, ships, tanker truck, and railways will be collectively required for a large scale $CO_2$ transport network. In particular, ship transportation can play a special role in the early phase of CCS infrastructure development as it offers flexible routes between source and offshore storage sites. This work is focused on the design and operational strategy of the $CO_2$ terminal which acts as a connecting link between $CO_2$ liquefaction and the shipping section. The study is performed using dynamic simulation in order to have a more realistic understanding of the process. Four scenarios have been developed to define the operational strategy of the terminal: loading case, holding case, unloading case, and emergency shutdown. A reasonable equipment sizing has been done after consultation with vendors data and available literature. The heat influxes to storage tanks include the heat leak from the bottom surface, dry wall, and wet wall for the precise quantification of heat flow across the system. Reasonable control loops are appended to keep the process variables at their set points to allow the safe operation of the $CO_2$ terminal within the operational limits. Finally, a sensitivity analysis has been performed on some of the important design variables in order to understand their effect on the process performance and terminal operation. The results show that boil-off gas (BOG) generation within the $CO_2$ terminal depends on storage tank size, operating pressure, ambient conditions, insulation thickness, and the filling level of the vessel.

## 1. INTRODUCTION

A new report by the Intergovernmental Panel on Climate Change (IPCC) shows that global emissions of greenhouse gases grew more quickly between 2000 and 2010 than in each of the three previous decades despite a growing number of policies to reduce climate change.[1] One of the many factors is the lack of attention by policy-makers and public to accept and implement CCS on a commercial level. This decade is critical for moving CCS through and beyond the demonstration phase.[2] The transport of captured $CO_2$ from the source sites to various geological locations is an important element for the large scale development and implementation of the CCS network. $CO_2$ can be transported by pipelines, ships, tanker trucks, and trains, depending on region, economics, regulation, location, and type of storage site. Currently, pipeline transport is mainly used in the US to transport over 50 million tons of $CO_2$ per year for enhanced oil recovery (EOR) projects. However, no such infrastructure exists in other parts of the world, and the construction of a dedicated pipeline network will require a considerable capital expenditure. Ship transportation, on the other hand, can play an important role in the early phase of CCS infrastructure development as it offers flexible routes between source and offshore storage sites. Also, foreseeing the uncertainty in the CCS market, ships represent a certain residual value, which means that in the case of CCS project cancellation, capital (CAPEX) invested will not be wasted all together and the ship can be employed in alternative trades.[3] Ship transportation itself can be divided into three subparts as shown in Figure 1. First, the captured $CO_2$ is liquefied through the liquefaction plant which is an integral part of ship transportation. The second part is the storage of liquefied $CO_2$ into intermediate storage tanks, until it is loaded on the ships. The third part can be seen as the transport of the $CO_2$ ship from the $CO_2$ terminal to the storage site.

Recently, many researchers have focused on various aspects of $CO_2$ ship transportation. Aspelund and Jordal[4] proposed a liquefaction process design for ship transport with liquefaction energy requirement of 105 kWh per tonne of $CO_2$. Lee et al.[5] modified the basic design proposed by Aspelund and Jordal[4] and suggested two alternative designs. This was done by making use of cold energy from flashed $CO_2$ streams and by optimization of the compression ratio. The results showed a reduced liquefaction energy consumption of 98.9 kWh per tonne of $CO_2$. More recently, Zahid et al.[3] proposed a liquefaction process design for postcombustion and precombustion source facilities and recommended an optimum location of liquefaction plant based on various scenarios. Chiyoda corporation[6] proposed the concept of shuttle ships which are equipped with injection facilities; hence, there is no need for an injection platform at the injection site. Alexandrakis and Smart[7] proposed the use of a $CO_2$ ship for multiproduct transport. For example, a $CO_2$ ship can be used to transport $CO_2$ to the storage site and then be used to transport another liquefied gas product available from the storage site or another location and return back to the $CO_2$ terminal. This concept looks attractive from the supply chain perspective, but it will make the system more complicated. Also, vessel cleaning will be required each time before the ship is loaded with a different type of liquefied product. Another study[8] considered insufficient land at the capture site and proposed a floating

**Received:** September 18, 2014
**Revised:** December 26, 2014
**Accepted:** February 5, 2015

ACS Publications © XXXX American Chemical Society



DOI: 10.1021/ie503696x
*Ind. Eng. Chem. Res.* XXXX, XXX, XXX−XXX

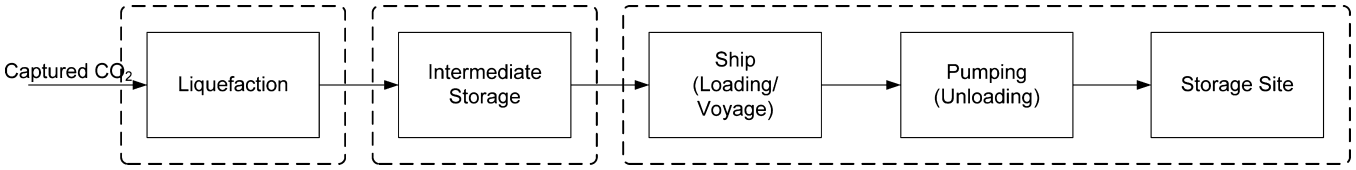



**Figure 1.** $CO_2$ shipping value chain.

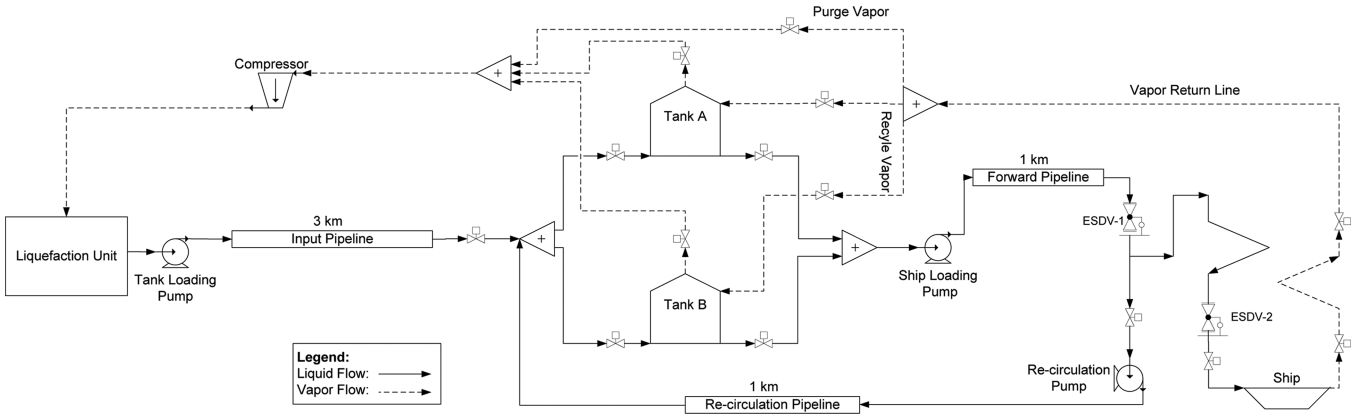



**Figure 2.** Process flow diagram of $CO_2$ terminal.

barge design for $CO_2$ liquefaction and intermediate storage. This type of floating barge system can have a single tug and multiple barges, with the tug accompanying one barge at a time.

Most of the studies addressing $CO_2$ shipping mentioned the need for intermediate $CO_2$ storage tanks and included it in economic analyses to account for the tank cost. However, shall the CCS technology move further toward commercialization, the $CO_2$ intermediate tank will not be simple buffer storage but rather will develop as a $CO_2$ terminal with huge volume influxes. The industry has a long experience operating and managing LNG terminals that may be used as guidance for the development of $CO_2$ terminals. However, the operating conditions of the $CO_2$ terminal will be quite different than that of the LNG terminal. LNG is stored and transported at atmospheric pressure, while $CO_2$ needs to be stored and transported at a minimum pressure of 6.5 bar. The cargo density of LNG is about 450 kg/m$^3$ compared to the $CO_2$ transportation density of 1100 kg/m$^3$. This means that $CO_2$ ships can carry more mass than an equivalent LNG ship. Liquid $CO_2$ in a storage tank or ship is not as cold as LNG, which is usually at −160 °C. Fire and explosion can result from loss of containment of LNG, while high concentration of $CO_2$ can cause personnel asphyxiation.

Lee et al.[9] presented the conceptual design of a $CO_2$ terminal with its operating conditions. The scope of the study included the $CO_2$ storage tank, BOG generation due to heat influx in the system, and its reliquefaction facility. The study also compared the BOG generation under good insulation and bad insulation conditions and consequently the liquefaction energy required to reliquefy it. Their study was based on the steady-state model built using a commercial simulator. However, the study was not able to explain the behavior of liquid $CO_2$ and its BOG generation during different stages of terminal operations like $CO_2$ filling in the tank, holding condition, and subsequently, $CO_2$ loading from the tank to the ship. Similar to the assumptions made for LNG terminals, this study assumed that the cryogenic conditions must be maintained within the recirculation line. However, the recirculation flow rate was calculated as 52% of the input flow which may contribute to an increased pumping energy. A knowledge sharing report for the Rotterdam CCS network[10] also discussed the operating conditions of the $CO_2$ terminal. This study assumed three different pressures of 7, 8, and 9 bara for liquefaction, intermediate storage, and shipping. The analysis showed that a storage pressure of 7 bara seems to be the optimum pressure for the overall chain. The study also performed a sensitivity analysis on the tank pressure by varying the heat influx to the storage tank.

Dynamic modeling, which is a collection of time-changing patterns to represent a process, is a useful tool for predicting and analyzing the transitions between cases. None of the previous studies investigated the dynamic and reliable operational strategy of $CO_2$ terminal, its BOG handling system, and control philosophy of the tanks. The objective of this study is to explore the operational issues of $CO_2$ terminal that require attention at the design stage. This study is mainly divided in two parts: design and operation of the $CO_2$ terminal. A dynamic simulation model is made in this study to investigate the various operational modes of the $CO_2$ terminal. Key system parameters, such as storage tank pressure, vessel liquid level, BOG flow rate, and recirculation flow rate are tracked during simulation to ensure the safe operation of the $CO_2$ terminal with a minimum energy requirement. Essential control schemes have been employed to provide good reliability and keep process variables at their set points. A detailed heat transfer calculation method is followed for the precise quantification of heat flux across the system.

## 2. TERMINAL DESIGN

The $CO_2$ terminal should be designed in a way such that it can be operated safely while keeping the $CO_2$ in liquid phase throughout the process. However, due to heat inleaks into the system, the production of BOG is inevitable. Figure 2 shows the process diagram for the $CO_2$ terminal. $CO_2$ is liquefied at the liquefaction plant and then transported through an insulated pipeline to an intermediate storage tank where it is


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

stored safely. The $CO_2$ terminal can have a number of storage tanks, depending on the liquefaction plant capacity and the shipping schedule. With dependence on the specific operation mode, a certain quantity of BOG is continuously produced from the storage tank, which is recycled back to the liquefaction plant. Once the ship arrives at the loading port, the liquid $CO_2$ is pumped from the storage tanks through the forward pipeline to the ship. A small part of the liquid $CO_2$ is recirculated through the recirculation pipeline so that when the $CO_2$ loading to the ship is started, low temperature can be maintained without vapor flashing in the forward pipeline.

The composition of the captured $CO_2$ stream may vary depending on the source and technology used for $CO_2$ capture. The variation of impurity content affects the vapor liquid equilibrium (VLE) properties of $CO_2$ mixtures, mainly boiling and condensing behaviors. The design and operation of the terminal can be influenced by changing the thermodynamic properties of $CO_2$ mixtures due to presence of impurities. Therefore, noncondensable gases should be removed from the captured $CO_2$ stream in order to satisfy the requirements of transportation and use the storage reservoir efficiently. An increase of noncondensable gases in the $CO_2$ stream raise the boiling and condensing pressure compared with the saturated state of pure $CO_2$. Particularly, the variation of $N_2$ has the most remarkable impacts on both the dew points and the bubble points of $CO_2$ mixtures.[11] Different from the behavior of the noncondensable gases, $SO_2$ has the opposite impact on the VLE properties of $CO_2$ mixtures. Since $SO_2$ has a higher critical point than $CO_2$, the presence of $SO_2$ in $CO_2$ mixtures will make the condensing temperature increase at a certain pressure or make the condensing pressure decrease at a certain temperature.[11] In this study, a pure $CO_2$ stream is assumed in order to have a basic understanding of the $CO_2$ terminal operations.

On the basis of the storage tank status, this study has divided the operation of the $CO_2$ terminal into three basic modes: loading mode, holding mode, and unloading mode. A special case of emergency shutdown (ESD) has been included in the event of an emergency situation. Table 1 shows the design parameters for the base case used in this study. This study assumes that liquid $CO_2$ is available from the liquefaction plant at −52 °C and 6.5 bar. During the loading mode, liquid $CO_2$ from the liquefaction plant is pumped to the intermediate storage tank where it is kept at a pressure of 7.0 bar. The liquid transferred to the tank results in a level build up in the storage vessel and compression of the vapor phase above the liquid. This vapor has to be removed from the tanks to prevent excessive pressure build-up in the tanks. A continuous vapor stream is removed from the storage tank and returned to the liquefaction plant during the loading mode. Once the tank is filled with liquid $CO_2$, it is considered to be in holding mode until the tank is unloaded. The safe liquid $CO_2$ hold up during the holding mode depends on the ambient temperature, tank insulation material, insulation thickness, and filling level of the tank.[10] The tank is under the unloading mode when the liquid $CO_2$ is pumped from the storage tank to the ship. The removal of liquid $CO_2$ from the storage tank will reduce the pressure which can eventually result in solidification of the tank content. Hence, there is a need to compensate for this pressure drop by feeding $CO_2$ vapor into the tank. For this purpose, the vapor generated while loading $CO_2$ into the ship will be returned from the ship back to the storage tanks. This will prevent pressure reduction in the storage tank and pressure build up in the ship. BOG load varies by a large degree between loading mode, holding mode, and unloading mode.

**Table 1. $CO_2$ Terminal Parameters for the Base Case Design**

| | |
|---|---|
| terminal annual capacity | 5 million ton $CO_2$ |
| $CO_2$ specification at liquefaction plant | 6.5 bar, −52.0 °C |
| $CO_2$ specification at storage tank | 7.0 bar, −48.8 °C |
| $CO_2$ specification at ship | 7.1 bar, −48.6 °C |
| storage tank capacity | 5000 $m^3$ |
| storage tank type | spherical |
| number of tanks | 2 |
| ambient temperature | 30 °C |
| insulation material | polyurethane |

Table 2 presents a methodology explaining how different design parameters should be selected for the basic design of a $CO_2$ terminal, what specific design codes should be followed in order to ensure the terminal safety and what process variables should be monitored during different operational modes. The table consists of sections including design basis, cryogenic equipment design, layout of the terminal, ship vessel sizing, and operational parameters that require attention for the safe operation of the terminal. The details for design and operation of the $CO_2$ terminal are explained further in the following sections.

**2.1. Tank Design.** One of the characteristics of $CO_2$ is the absence of a normal boiling point which means that $CO_2$ can only exist as gas or solid at an atmospheric pressure. $CO_2$ in the liquid phase can only exist above the triple point pressure of 5.2 bar. However, if the storage pressure is increased significantly, it will cause an increase in CAPEX. Therefore, the operating pressure of the $CO_2$ terminal is set to be in the range from 6.5 to 7.5 bar with the normal operating pressure of 7.0 bar for the storage tank and 7.1 bar for the ship vessel. In this way, a safety margin can be maintained to avoid any dry ice formation below the triple point and the $CO_2$ tank will cost less compared to current shipping practices at higher pressure. In this study, spherical storage tanks with a maximum design pressure of 9 bar are used. Absence of corners result in the even distribution of stresses on the sphere's surfaces, both internally and externally which adds to its strength. Moreover, a sphere has smaller surface area per unit volume compared to other vessel shapes. This means that the heat flux from the outside environment to the liquid $CO_2$ will be less than that for cylindrical or other storage vessels. The design and safety control loops for the $CO_2$ storage tank are shown in Figure 3. The thickness of the tank material is calculated using the ASME code section VIII (Pressure Vessels) for hemispherical heads as shown in eq 1.[12]

$$t = \frac{PL}{2SE - 0.2P} \tag{1}$$

where $t$ is the minimum thickness of the tank, $P$ is the maximum internal design pressure, $L$ is the inside radius, $S$ is the allowable stress in tension, and $E$ is the joint quality factor. The allowable stress $S$ value is taken as 129.2 MPa for steel tanks and $E$ is set to 1.0, assuming fully radiographed tanks.[13] ASTM A203 has been followed for the compliance of $CO_2$ tank material. The material used is 3.5% nickel alloy steel that can be used at temperatures down to −100 °C. Safety factor represents the structural capacity of a system beyond the expected load and is a ratio of tensile strength to the yield point. The safety factor for 3.5% Ni-steel used in this study is 1.75.[14]


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

Table 2. Scheme for the Basic Design of $CO_2$ Terminal

| design parameter | description |
|---|---|
| a. design basis | |
| • determine the terminal throughput capacity | • based on the $CO_2$ demand for the permanent storage or utilization |
| • define operating condition | • select suitable temperature and pressure based on thermodynamic analysis |
| • number of storage tanks required | • based on terminal throughput for discrete operation (also involves maximum buildable size based on material selection) |
| b. terminal conceptual design | |
| • determine required equipment | • must be decided based on required performance while keeping high reliability, operability, and maintainability |
| • develop process flow sheet | • based on the starting and stopping points of the process |
| **1b. tank design** | • ASME code section VIII, division 1 (pressure vessels) |
| • tank material selection | • ASTM standards (for cryogenic applications) |
| • tank type selection | • based on previous industry experience |
| • maximum design pressure | • 20% × maximum operating pressure + maximum operating pressure |
| • tank thickness | • ASME code section VIII, division 1 (pressure vessels) |
| • storage tank sizing | • based on liquid hold-up requirement |
| **2b. pipeline design** | • ASME B31.4 (pipeline transportation systems for liquid hydrocarbons and other liquids) |
| • pipeline material selection | • ASTM standards (for cryogenic applications) |
| • pipeline diameter calculation | • based on optimal pipeline diameter from Design and Economics for Chemical Engineers |
| • pipeline thickness | • ASME B31.4 (pipeline transportation systems for liquid hydrocarbons and other liquids) |
| **3b. insulation material** | |
| • selection of suitable insulation material | • insulation for cryogenic equipments (refer to ref 13) |
| • insulation thickness | • based on allowable boil-off rate (BOR) and economic comparison between capital and operating cost |
| b. terminal conceptual design | |
| **4b. heat flux** | |
| • calculate heat flux across storage tanks | • refer to ref 18 |
| • calculate heat flux across pipeline | • refer to eq 7 and 8 |
| c. layout of the terminal | |
| • location of the equipment | • facilities are laid out according to the sequence of processes at a terminal |
| • determine pipe lengths based on layout | • determine process and utilities pipeline lengths based on the equipment placement |
| d. ship vessel sizing | |
| | • based on transport capacity, transport distance, design speeds of the ship (laden and ballast), loading and unloading time requirement, and number of ships used |
| e. terminal operation | |
| **1e. loading mode** | • monitor BOG generation in the tank as per design |
| | • pressure controller operation and its tuning for reliable control |
| | • level controller operation and its tuning for reliable control |
| | • calculate minimum recirculation flow rate during loading mode |
| **2e. holding mode** | • ensure proper working of pressure controller |
| **3e. unloading mode** | • monitor BOG generation in the ship vessel |
| | • ship vessel pressure controller operation and its tuning for reliable control |
| | • ship vessel level controller operation and its tuning for reliable control |
| | • tank pressure controller operation and its tuning for reliable control |
| | • calculate minimum recirculation flow rate during unloading mode |
| | • calculate recycle vapor streamflow rate from ship to tank |

**2.2. Pipeline Design.** As shown in Figure 2, four different pipelines are part of the $CO_2$ terminal. These include the input pipeline from liquefaction plant to storage tank, the forward line, the recirculation line, and the vapor return line. Equation 2 proposed by Zhang el al.[15] is used to calculate the optimal pipeline diameter.

$$D_{opt} = 0.363\left(\frac{Q_m}{\rho}\right)^{0.45}\rho^{0.13}\mu^{0.0025} \tag{2}$$

where $D_{opt}$, $Q_m$, $\rho$, and $\mu$ are the optimal pipe diameter ($m$), mass flow rate (kg/s), $CO_2$ density (kg/m$^3$), and viscosity (Pa s), respectively. This equation was originally derived by Peters[16] for the calculation of economic pipe diameter, assuming average numerical values for some of the less critical parameters and has been used for pipeline diameter calculations in this study. The Beggs and Brill[17] method has been used to calculate the pressure drop through the pipelines within the $CO_2$ terminal. This correlation was the first developed method to apply to all pipe inclination angles and has a good applicability for different pipeline geometries. The total pressure drop through the pipeline is the sum of pressure drops due to potential energy change, kinetic energy change, and friction losses as shown in eq 3.

$$\left(\frac{dp}{dL}\right) = \left(\frac{dp}{dL}\right)_{PE} + \left(\frac{dp}{dL}\right)_{KE} + \left(\frac{dp}{dL}\right)_{fric} \tag{3}$$

However, since there is no elevation change involved in this case, the pressure drop due to potential energy can be neglected. The pressure drop caused due to kinetic energy is proportional to the change in the fluid velocity and is given by eq 4:

$$\left(\frac{dp}{dL}\right)_{KE} = \rho v\left(\frac{dv}{dL}\right) \tag{4}$$

The pressure drop across the pipeline depends mainly on the frictional pressure gradient and is calculated using eq 5 as shown below:

$$\left(\frac{dp}{dL}\right)_{fric} = \frac{f\rho v^2}{2g_c D} \tag{5}$$



DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

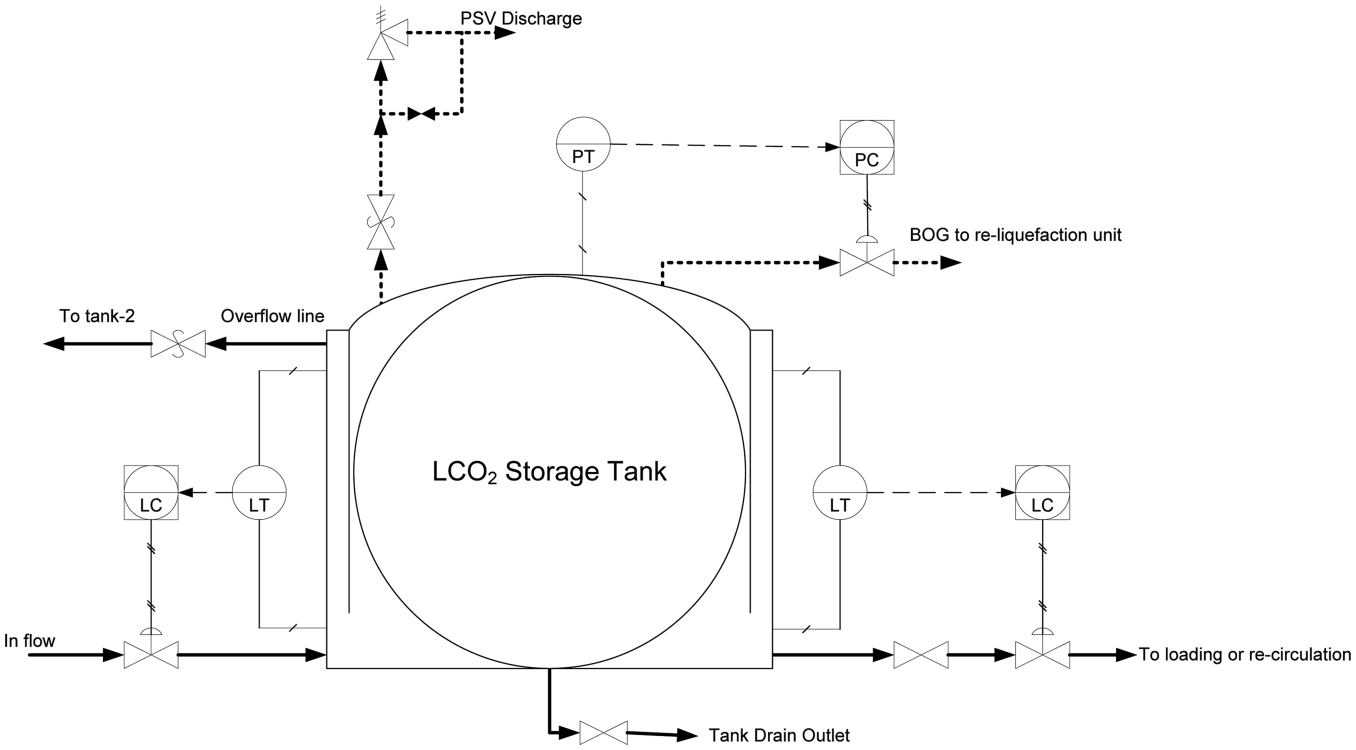



**Figure 3.** Design and control of $CO_2$ storage tank.

where $f$ is the friction factor, $\rho$ is the fluid density (kg/m$^3$), $v$ is the in situ velocity (m/s), $L$ is the length of pipe (m), $g_c$ is the conversion factor (kg m/N s$^2$), and $D$ is the pipeline diameter (m). The fluid velocity ($v$) is calculated on the basis of the volumetric flow rate and calculated pipeline diameter, whereas the friction factor is calculated based on the Reynolds number and relative roughness of the pipeline. The calculated diameter and pressure drop for different pipelines are presented in Table 3. The pressure drop across the forward and recirculation line increases significantly during the unloading mode than that during the loading mode operation.

**Table 3. Calculated Pipeline Diameters**

| | nominal pipe size (in.) | pressure drop (kPa) |
|---|---|---|
| input pipeline | 12 | 277 |
| forward pipeline (during loading mode) | 14 | 5.1 |
| forward pipeline (during unloading mode) | | 150 |
| recirculation pipeline (during loading mode) | 10 | 11 |
| recirculation pipeline (during unloading mode) | | 44 |
| vapor return line | 3 | – |

**2.3. Heat Flux Calculation.** The total heat flux is calculated across the pipelines and storage tanks to precisely determine the BOG load which arises from the heat transfer using eq 6.

$$dQ = dQ_{pipe} + dQ_{tank} \tag{6}$$

where $dQ$ is the total heat flux across the system, $dQ_{pipe}$ is the heat flux over the pipelines, and $dQ_{tank}$ represents the heat transfer across the storage tanks. $dQ_{pipe}$ can be calculated by using eqs 7 and 8.

$$dQ_{pipe} = U_{o,pipe}\Delta T\, dA \tag{7}$$

$$U_{o,pipe} = \frac{1}{r_o\left(\frac{1}{r_o h_o} + \frac{\ln(r_1/r_o)}{k_{01}} + \frac{\ln(r_2/r_1)}{k_{12}} + \frac{1}{r_2 h_2}\right)} \tag{8}$$

where $U_{o,pipe}$, $h$, and $k$ represent the overall heat transfer coefficient, convective heat transfer coefficient, and thermal conductivity, respectively, and $r_0$, $r_1$, and $r_2$ represent the internal radius, external radius, and outside radius of insulation for the pipeline. The heat flux across the storage tank is calculated using a procedure outlined by Kumana and Kothari[18] in which the heat fluxes are calculated separately for each category of surfaces. In the case of a spherical tank, these surfaces are wet wall, dry wall, and tank bottom. Wet wall refers to the portion of the wall submerged under the liquid surface, whereas dry wall refers to the portion of the wall in the vapor space, above the liquid surface. In this study, heat transfer calculations are based on the maximum tank filling level of 93%, which is a recommended maximum fill level from vendors' data.[19] $dQ_{tank}$ can be calculated using the following eqs 9–12.

$$dQ_{tank} = dQ_{wet} + dQ_{dry} + dQ_{bottom} \tag{9}$$

$$dQ_{wet} = U_w\Delta T\, dA \tag{10}$$

$$dQ_{dry} = U_d\Delta T\, dA \tag{11}$$

$$dQ_{bottom} = U_b\Delta T\, dA \tag{12}$$

Where $dQ_{wet}$, $dQ_{dry}$ and $dQ_{bottom}$ represent the heat transfer through wet wall, dry wall, and bottom surfaces, respectively. $U_d$, $U_w$, and $U_b$ are the overall heat transfer coefficients for the dry wall, wet wall, and bottom surfaces, respectively. It may be difficult to estimate the overall heat transfer coefficients $U_d$, $U_w$, and $U_b$ for the different surfaces of the tank. For the tank


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

geometry chosen in this study, these can fortunately be calculated using the published correlations.[20] The heat transfer coefficient for air is taken as 50 W/m$^2$ K for wind speeds of up to 15 mph. Table 4 shows the calculated heat fluxes for the

**Table 4. Heat Flux Across the $CO_2$ Terminal**

| | heat influx (kW) |
|---|---|
| storage tank | 31.7 |
| ship vessel | 48.0 |
| input pipeline | 84.4 |
| forward pipeline (during loading mode) | 28.0 |
| forward pipeline (during unloading mode) | 32.5 |
| recirculation pipeline (during loading mode) | 27.9 |
| recirculation pipeline (during unloading mode) | 27.9 |

pipelines and storage tank. The result shows that the heat transfer across the forward line increases during the unloading mode because of much higher flow rate compared to that during the loading mode.

## 3. TERMINAL OPERATIONAL DYNAMICS

The process performance of the $CO_2$ terminal under different operational modes cannot be described clearly by a static analysis. Therefore, a dynamic simulation has been performed using the commercially available simulation software Aspen HYSYS. The simulations are done using Soave−Redlich−Kwong (SRK) as the physical property package, since it can precisely predict the phase behavior of $CO_2$. The equipment consists of a tank loading pump, pipeline from liquefaction plant to the $CO_2$ terminal, storage tanks, a ship loading pump, recirculation pump, forward pipeline, and recirculation pipeline, as shown in Figure 2. The tank loading pump sends the liquefied $CO_2$ through a 3 km long pipeline from the liquefaction plant to the $CO_2$ terminal where intermediate storage tanks are located. The heat leaks from the environment to the system continuously generate BOG. Therefore, in order to reduce BOG, the pipelines and storage tanks are insulated with polyurethane material with a thermal conductivity of 0.018 W/m K. The minimum recirculation flow of $CO_2$ required to maintain cryogenic conditions within the pipelines is sent through the forward pipeline back to the storage tank. The forward line and recirculation line each are assumed to have a length of 1 km.

**3.1. Loading Mode.** Liquid $CO_2$ available at −52 °C and 6.5 bar is pumped to the $CO_2$ terminal where it is stored in a semipressurized storage tanks. The tank is operated with a set point pressure of 7.0 bar. The quantity of vapor in the tank outlet increases significantly as soon the $CO_2$ loading starts in the tank. These vapors are a combination of volume displaced in the tanks by the incoming $CO_2$, vapor resulting from the pumping energy in the pipeline, and vaporization resulting from heat leaks through the insulated tank and pipelines. The formation of $CO_2$ vapors results in excessive pressure build up in the tank. Therefore, during the loading mode, a continuous vapor flow is removed from the storage tank in order to maintain the vessel pressure. While the $CO_2$ is filling in the tank, a minimum liquid flow rate from the tank is pumped through the forward line and recirculation line to keep the $CO_2$ terminal within cryogenic operating limits.

In this study, $CO_2$ loading in the intermediate storage has been performed with a tank capacity of 5000 m$^3$. Initially, the storage tank is made empty with no liquid $CO_2$ and BOG in it. Also, before the $CO_2$ loading starts, the temperature of forward and recirculation pipelines are assumed to be at ambient temperature. Figure 4 shows the BOG generation in the 5000

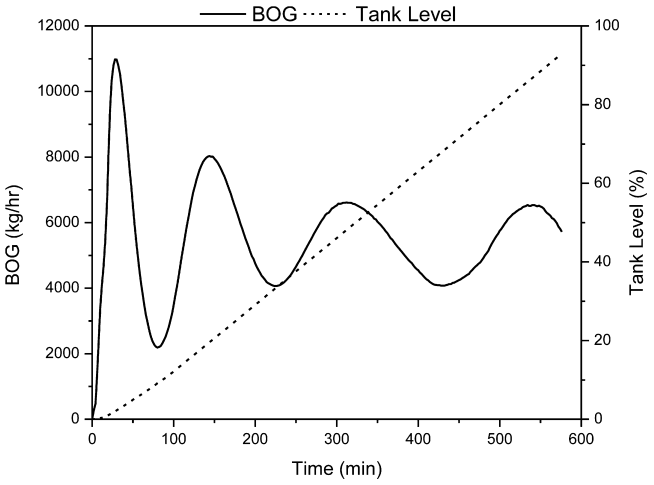



**Figure 4.** BOG generation and tank level during 5000 m$^3$ tank loading mode.

m$^3$ capacity tank during the loading mode. As soon as the liquid $CO_2$ enters the tank when starting to fill the tank, it flashes to the vapor phase and a huge amount of $CO_2$ vapors are generated. As the tank loading continues, the vapor and liquid remain in equilibrium with each other. The liquid $CO_2$ exists in equilibrium with its vapor inside the tank with the tank pressure being dependent solely on the temperature. The high level switch of storage tank is set at 93% liquid volume percent of the tank which is controlled by the level controller. The results show that it takes about 9.6 h to fill 93% of a 5000 m$^3$ $CO_2$ tank. Figure 5 shows the pressure control of the tank during the

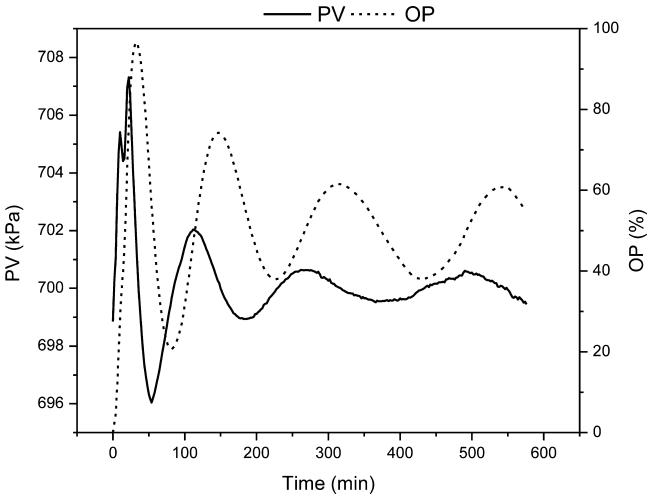



**Figure 5.** Tank pressure control behavior during loading mode.

loading mode. As a large quantity of BOG (11000 kg/h) is generated in the start of tank filling, the pressure of the tank increases from a set point of 700 kPa to around 707.3 kPa. To keep the tank pressure at its set point, BOG is drawn out from the top of the storage tank. The pressure controller used in this study is a simple PI controller given with a set point of 700 kPa. The tuning parameters for the pressure PI controller are given



DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

in Table 5. The gain ($K_c$) and integral time ($T_i$) values are selected using the rules of thumb values with autotuning done within the simulator. The valve opening percentage (OP) follows closely with the process variable (PV), which is the tank pressure, to keep it at the set point.

**Table 5. Tuning Parameters for the PI Controllers**

| | gain ($K_c$) | integral time ($T_i$) |
|---|---|---|
| tank pressure controller | 2 | 3 |
| ship pressure controller | 2 | 3 |
| recirculation flow rate controller | 0.5 | 0.3 |

As mentioned previously, a minimum liquid flow rate needs to be recirculated through the terminal to keep pipelines within the cryogenic limits. During the tank loading mode, the liquid $CO_2$ recirculation flow rate depends on the heat flux through the system caused by pumping, turbulent flow, and line frictions. Any heat leaking into the system will increase the temperature of the liquid $CO_2$ and form additional vapors in the tank. The quantity of these additional vapors depends on the length of the pipeline (forward and recirculation) and the power of the pumps. The vapor formed in the storage tank can be taken out as the BOG stream. However, any significant vapor forming in the pipeline is undesirable as it can lead to safety issues such as pipeline hammering and pump cavitations. The liquid $CO_2$ recirculation flow rate is calculated based on the presence of any $CO_2$ vapor in the recirculation line. Recirculation flow rate is calculated so that no vapor is formed in the pipeline. This is done by installing a PI controller with a given set point of zero vapor fraction in the recirculation pipeline. The controller takes the input by measuring the vapor fraction in the recirculation line and manipulates the recirculation pump power. The tuning parameters for this PI controller are given in Table 5. The calculated recirculation flow rate ensuring no vapor formation is around 113000 kg/h, which is about 19.5% of the input flow rate. As long as the operating temperature of the terminal remains within the cryogenic limits, the minimum recirculation flow rate is pumped to keep the operating energy as low as possible.

**3.2. Holding Mode.** The holding mode refers to a state after liquid $CO_2$ is filled in the tank and no liquid is being

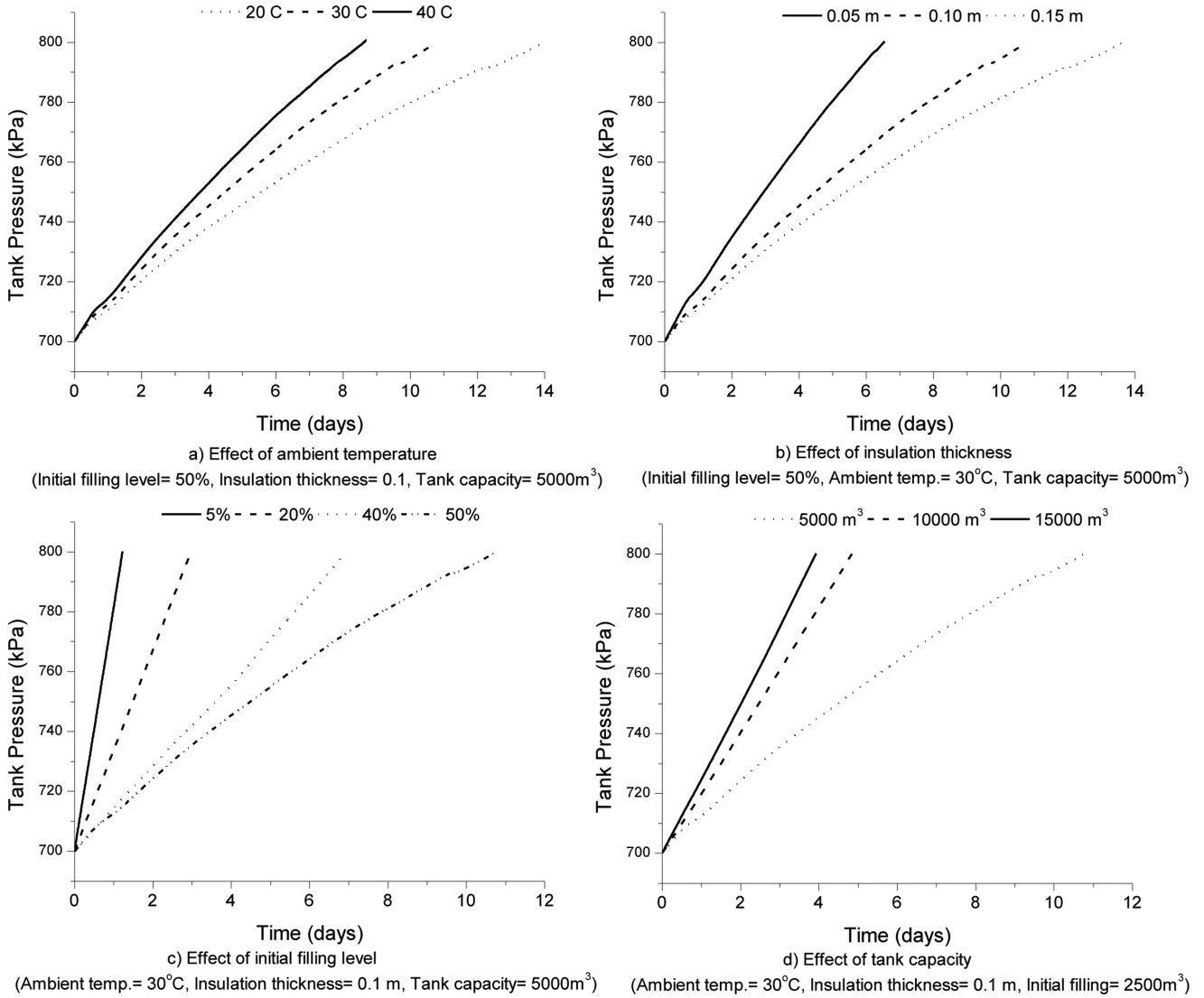



**Figure 6.** Effect of (a) ambient temp; (b) insulation thickness; (c) initial filling level; and (d) tank capacity on the pressure build up when BOG is not withdrawn from the tank during the holding mode.



DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

withdrawn from it. The liquid $CO_2$ can be stored in the tank for long durations, as long as heat leaks into the tank are minimized. Decarre et al.[21] studied the $CO_2$ ship transportation and assumed that the BOG shall neither be released to the atmosphere nor reliquefied during ship transportation. They reported a possible pressure increase in the vessel from 7 to 10.4 bar for a transport distance of 100 km. In such a case, when the BOG stream is not taken out from the tank, the pressure of the tank rises. The design pressure of the tank limits the amount of BOG that can be handled without releasing or reliquefying it. However, heat flux through the tank depends on many factors like ambient temperature, tank insulation material, insulation thickness, and filling level of the tank. In the case of LNG, BOG generation during the holding mode is 8−10 times less than during the loading mode.[22] Under specific site conditions and insulation material, $CO_2$ hold up time in the storage tank can only be increased by using a tank with higher design pressure, which results in an increase in the cost of the vessel. In this study, dynamic simulation studies are done to investigate the effect of ambient temperature, insulation thickness, filling level of the tank, and tank capacity on the pressure build up in the tank when BOG is not withdrawn during the holding mode.

*3.2.1. Effect of Ambient Temperature.* The ambient temperature is a site specific parameter that can differ significantly from one region to another. Even for one specific location, the ambient temperature may vary throughout the year and an average temperature difference between winter and summer can be as high as 15 °C. Figure 6a shows the effect of ambient temperature on the pressure build up in the storage tank. The results show that a high ambient temperature will result in more heat transfer through the tank, which in turn will produce more BOG. Hence, $CO_2$ terminals located in the cold climate regions will have a natural advantage in terms of operational energy penalty. The storage tank located in colder ambient temperature regions can hold liquid $CO_2$ for much longer periods of time compared to those located in hotter climate.

*3.2.2. Effect of Insulation Thickness.* The heat flux through the system depends on the type of insulation used and its thickness. The higher the thermal resistivity of insulation, the lower the amount of heat that will leak into the system. The thermal resistance of the insulation material can be increased by increasing the insulating layer thickness. In this study, polyurethane foam has been used as an insulation material with thermal conductivity of 0.018 W/m K. The insulation thickness is a trade-off between insulation costs and energy required to reliquefy the generated BOG. However, the optimization between insulation thickness and reliquefaction energy is not in the scope of this study. Throughout this study, an insulation thickness of 0.1 m is used and a sensitivity analysis is done to show the effect of insulation thickness on the pressure build up in the tank. Figure 6b shows that increasing the insulation thickness decreases the BOG generation in the tank, and hence, liquid $CO_2$ can be stored for a longer period of time.

*3.2.3. Effect of Tank Filling Level.* Tank filling level is an operational and logistic parameter that is an important factor in the generation of BOG.[10] As described earlier, the maximum filling level of the $CO_2$ tank is set at 93%. However, depending on the liquefaction plant operation and $CO_2$ shipping logistics, the available $CO_2$ level in the tank may vary from day-to-day operations. For example, depending on the size of ship used for transportation of $CO_2$, a certain quantity of liquid $CO_2$ may be left in the tank after loading $CO_2$ to the ship. It is important from the operational point of view to monitor the $CO_2$ level remaining in the tank, as it can affect the operational energy requirements. Figure 6c shows the pressure build up for various $CO_2$ filling levels in the tank. A low level of liquid $CO_2$ in the tank can vaporize faster compared to a high filling level. It is thus recommended not to leave small quantities of liquid $CO_2$ in the tank. Any liquid $CO_2$ left in the tank due to operational or logistical lags should either be pumped to other tanks with high filling level or new liquid $CO_2$ production should be filled in that tank.

*3.2.4. Effect of Storage Tank Capacity.* The effect of storage tank's capacity on the pressure build up in the tank is presented in Figure 6d. The results show the same conclusion as discussed in Effect of Tank Filling Level. A certain quantity of liquid $CO_2$ present in different capacity tanks can be safely stored for a different period of time. This means that a fixed quantity of $CO_2$ can be stored longer in a small capacity tank compared to a larger capacity tank.

The results are useful for the careful management of the liquid $CO_2$ supply chain. Also, these results can be used as a guideline to define the maintenance schedule of the liquefaction plant.

**3.3. Unloading Mode.** The unloading mode refers to the pumping of liquid $CO_2$ from the storage tank to the ship. Once the ship arrives at the $CO_2$ terminal and is ready to be loaded, liquid $CO_2$ from the storage tank is pumped through the forward pipeline to the ship. The process of ship loading is similar to that of the $CO_2$ tank loading, as discussed previously in Loading Mode. However, there are some differences in terms of vessel operational conditions. When the ship loading pump starts to send liquid $CO_2$ to the ship, a large quantity of BOG is produced and taken out from the top of the vessel. The BOG produced during the storage tank filling is sent to the liquefaction plant for reliquefaction. However, the BOG produced from the ship is recycled back to the storage tank. The removal of the liquid $CO_2$ from the storage tank decreases the pressure in the tank. Hence there is a need to compensate for the pressure drop in the tank by recycling the BOG produced from the ship. In this way, the recycled BOG keeps the pressure of the ship and storage tank within operating limits. In this study, the ship vessel pressure is set at 7.1 bar, which is slightly higher than that of the storage tank. In the LNG terminals, a vapor return compressor or blower is usually installed to send the BOG from the ship to the storage tank. However, by keeping the ship vessel pressure slightly higher, additional operating and capital cost of such auxiliary equipment can be saved. The recirculation pipeline is kept within cryogenic conditions by sending a part of the liquid $CO_2$ flow through the recirculation pump back to the storage tank. However, the recirculation flow rate during the unloading mode is almost 2.4 times higher than that during the loading mode. Since during the unloading mode, liquid $CO_2$ is pumped from the storage tank to the ship and the recirculation line, without fresh inflow of $CO_2$ from the liquefaction plant, the recirculation flow rate is higher to accommodate the same quantity of heat flux as in Loading Mode. The storage tanks are not completely emptied during the unloading mode because the frequent start-up of tanks can decrease the strength of tank material and are not recommended from an operational perspective. Therefore, the minimum level of liquid $CO_2$ in the storage tank is set at 2.5% liquid volume percent of the


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

tank. The actual volumetric flow rate from the storage tank to the ship is set at 750 $m^3$/h. The ship capacity is calculated based on the simple shipping schedule. Four ships are assumed for each voyage route of 500 km. The design speed of the $CO_2$ ship is 15.0 knots in the laden voyage and 16.0 knots in the ballast voyage. The voyage time is calculated assuming 12 h of loading, 12 h of unloading, and 8 h of harbor/mooring time.[23] Hence, the time required for a single voyage is around 2.75 days for a 500 km trip, and each ship can make 122 trips per year. The high level switch of ship vessel is set at 95% liquid volume percent of the vessel. Therefore, the capacity of the ship is calculated as 9500 $m^3$ for an annual transport capacity of 5 Mton. Figure 7 shows the BOG generation in the 9500 $m^3$ ship

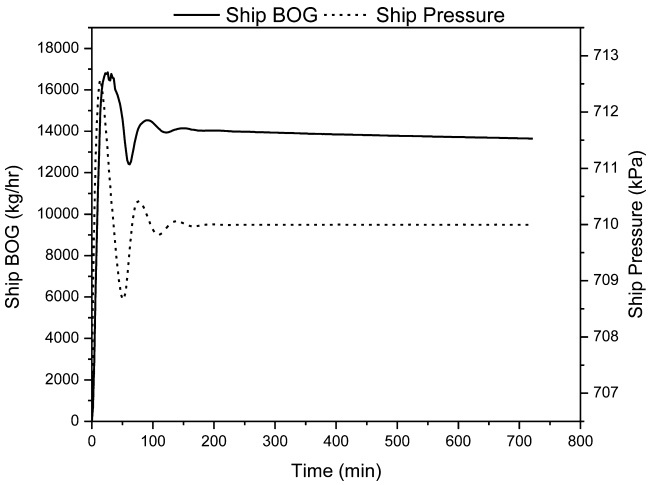



**Figure 7.** BOG generation and pressure variation in the ship during unloading mode.

along its pressure variation during the unloading mode. As soon as the liquid $CO_2$ enters the ship vessel, it flashes to vapor phase and a huge amount of $CO_2$ vapor is generated. The BOG generation from the ship stabilizes after 2 h of filling and remains constant at a flow rate of around 14000 kg/h. The ship vessel pressure is maintained within operating limits by using a pressure controller with tuning parameters shown in Table 5. It is evident from the results that as the ship vessel pressure changes, the corresponding BOG generated also change. BOG produced from the ship vessel is split up into two streams, recycle BOG and purge BOG. Figure 8 shows the flow rate of recycle and purge BOG during the unloading mode. The storage tank pressure is maintained at 7.0 bar by using a pressure controller, which manipulates the amount of recycle BOG back to the storage tank. The purge BOG stream is sent back to the liquefaction plant for reliquefaction. It can be seen from the results that most of the BOG generated from the ship is recycled back to the storage tank.

**3.4. Emergency Shutdown System.** Safe operation is the most important factor in the design of the $CO_2$ terminal. In order to protect personnel and prevent equipment damage, terminals are equipped with an emergency shutdown (ESD) system. The basic purpose of this system is to safely shut flows between ship and terminal in the event of an emergency. ESD system considered for the $CO_2$ terminal is similar in philosophy to systems applied for LNG terminals. ESD system usually includes fast closing valves, loading arms equipped with emergency release couplers, and sometimes, surge protection equipment. The ESD system may be activated automatically

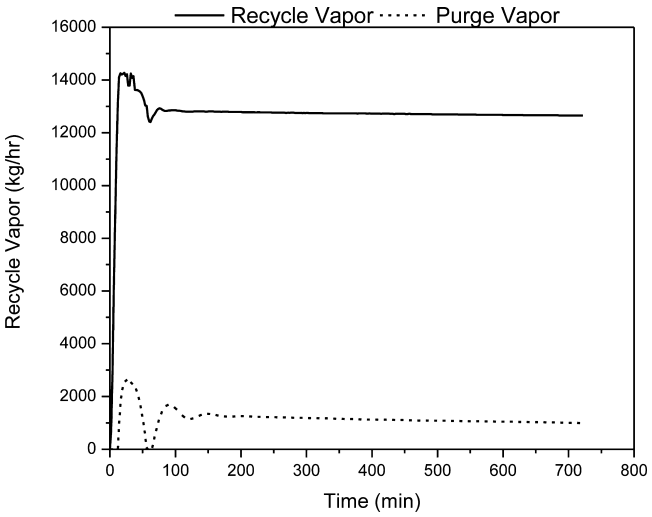



**Figure 8.** Flow rate of recycle and purge vapor from the ship during unloading mode.

when predefined operational parameters exceed their normal limit or it can be operator initiated. The cause of ESD system initiation can be an abnormal storage tank pressure or tank level, an abnormal ship vessel pressure or level, $CO_2$ leakage at terminal or ship, and loss of electrical power. Depending on the cause of ESD initiation, a hierarchy of actions and timing of valve closures must be developed.

In this study, a logical scheme of ESD system has been developed and its performance is analyzed by using the event scheduler function in the Aspen HYSYS. Two cases, namely ESD initiation during unloading mode and ESD initiation during loading mode, have been studied to analyze the ESD system response. During the unloading mode, the volume flow rate through the loading arms is quite high. Therefore, the first step in the case of ESD initiation is the stoppage of the ship loading pump that transfers liquid $CO_2$ from the tank to the ship along with the tripping of BOG compressor or blower located on the vapor return line. The next step is the closure of terminal and ship ESD valves to shut down the $CO_2$ flow across the terminal to the ship. This study assumed the closing time of ESD valves to be 30 s after the ESD is initiated. The response of ESD system during the unloading mode has been studied by activating the ESD system when the $CO_2$ level in the ship vessel was 40%. As soon as the ESD system was initiated, ship loading, and recirculation pumps were switched off. ESD valves, namely ESDV-1 and ESDV-2, were closed after 30 s of ESD initiation. Figure 9 shows the mass flow rates across the terminal before and after the ESD initiation. The liquid $CO_2$ flow rate through the forward pipe and recirculation pipeline was reduced to zero soon after the ESD activation. However, the return vapor flow rate from the ship took a little longer to reduce to zero because of the already present $CO_2$ vapors maintaining the ship vessel pressure. A shutdown time of 30 min is assumed in this study. Soon after the shutdown was resolved, the ESD valves were opened along with the pumps start up. Some quantities of $CO_2$ vapors were produced in the pipelines during the shutdown time. However, since the liquid recirculation flow rate is based on the presence of any $CO_2$ vapor in the recirculation line, fluctuation can be seen in the flow rates after the start-up time. Once the vapor fraction in the recirculation line was reduced to zero, the flow rates became steady after a 40 min start-up. The


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

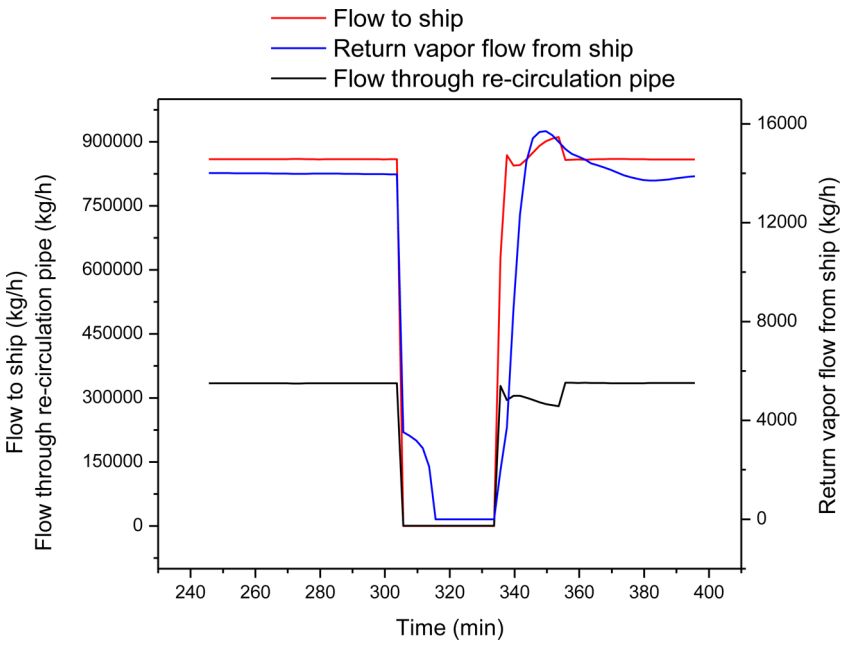



**Figure 9.** Flow rate variation for the period of ESD initiation during the unloading mode.

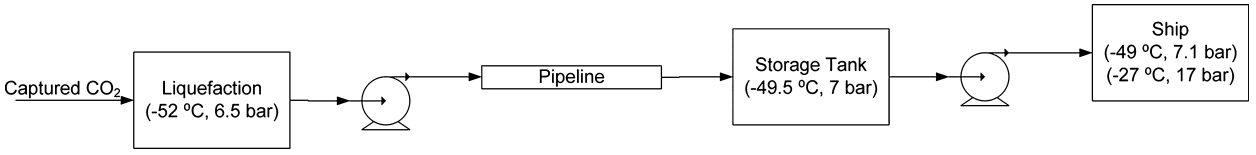



**Figure 10.** Conceptual diagram of dual operational terminal.

analysis shows that the valves and pipelines have been properly sized along with reasonable control loops.

During the loading mode, the ESD system may be activated due to a technical problem occurring at the liquefaction unit, storage tank high level, or abnormal pressure. The ESD system in response shuts down the tank loading pump, ship loading pump, and recirculation pump along with the closure of the ESD valve located at the input pipeline. ESD system during the loading mode showed similar results, as discussed above.

**3.5. Concept of Dual Operation Terminal.** Many studies reported that $CO_2$ ship transportation should be done near the $CO_2$ triple point, as discussed earlier. However, there is currently no commercial scale $CO_2$ plant operating at −52 °C and 6.5 bar. The existing commercial $CO_2$ storage and transportation practices are done at −27 °C and 17 bar. Since there is a need to demonstrate the economic viability of $CO_2$ ship-based transportation before large-scale implementations, shipping could be done at 17 bar in the initial phase using current ships, while new ships are developed to transport $CO_2$ at 7.1 bar for the larger capacities transportation. Therefore, current small capacity $CO_2$ ships and LPG ships may be deployed for the early phase demonstration projects. Hence, there is a need to design the $CO_2$ terminal in such a way that it facilitates the transition from the demonstration phase using current ships to the commercial phase using new ships at 7.1 bar. The $CO_2$ terminal design presented in this study is based on the concept of keeping the initial transition phase requirement in view. A simple process flow diagram of a dual operation terminal is shown in Figure 10. The liquefaction plant producing the liquid $CO_2$ operates at 6.5 bar, and the liquid $CO_2$ is pumped to the storage tank for storage at 7.0 bar. Depending on the operational parameters of the $CO_2$ ship vessel, the liquid $CO_2$ can be pumped to the ship either at 7.1 bar or at 17 bar. In this way, the current design of the $CO_2$ terminal can satisfy $CO_2$ shipping requirements, operating at different thermodynamic conditions.

## 4. DESIGN TRADE-OFFS

The $CO_2$ terminal acts as a connecting link between the $CO_2$ source and the sink. The objective of designing such a terminal is to ensure a reliable, economical, and safe operation while transporting $CO_2$ in and out of the terminal. Since, the $CO_2$ transport is always done under pressure, selection of optimum thermodynamic conditions is a key variable in the design of the $CO_2$ terminal. The pressure at which the $CO_2$ is liquefied, stored, and shipped presents the trade-offs between the capital and operating cost of the entire plant. A brief comparison is made to discuss the trade-offs between CAPEX and OPEX for different sections of the $CO_2$ terminal.

**4.1. Liquefaction Unit.** The OPEX slightly increases with an increase in the operating pressure of the liquefaction unit. However, operating the liquefaction unit at a higher pressure will have lower CAPEX compared to that operating at a lower pressure. This increase in CAPEX at a low liquefier pressure can be explained by the constraint of having a constant $CO_2$ yield for the liquefaction unit that requires additional pieces of equipment.

**4.2. Storage Tank.** An increase in the storage pressure will lead to a higher thickness of the tank, which causes an increase in the CAPEX. However, as will be discussed in Figure 12, an increase in the storage tank pressure reduces the amount of BOG generated that can help to decrease the OPEX.


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

Table 6. Cost Comparison at Different Operating Pressures[10]

| $CO_2$ pressure | liquefaction additional CAPEX (M€) | storage tank additional CAPEX (M€) | shipping additional CAPEX (M€) | overall additional CAPEX (M€) |
|---|---|---|---|---|
| 7 bara | reference case | | | |
| 8 bara | −1.7 | +5.9 | +0.8 | +5.0 |
| 9 bara | −2.3 | +11.9 | +1.5 | +11.1 |

**4.3. Shipping.** Ship vessel pressure has the same trade-off between CAPEX and OPEX as for the storage tank. Increasing the ship vessel pressure will increase the vessel thickness, causing an increase in the CAPEX. On the other hand, BOG generation will decrease with a pressure increase, hence a decrease in the BOG handling cost.

Table 6, adapted from a knowledge sharing report for the Rotterdam CCS network,[10] shows the impact of pressure on the CAPEX, taking 7 bar as a reference case. The analysis clearly shows that a storage pressure of 7 bar seems to be the optimum pressure for the overall chain.

## 5. SENSITIVITY ANALYSIS

The design presented in this study explains the operational behavior of a $CO_2$ terminal under different operational modes. However, a number of assumptions have been made for some of the design parameters, which may vary depending on the liquefaction plant design, terminal location and layout, $CO_2$ transport capacity, equipment design, and so on. Therefore, in order to estimate the influence of some of the assumed parameters, a sensitivity analysis has been performed on some of the important design variables.

**5.1. Effect of Storage Tank Size.** The base case assumed a storage tank capacity of 5000 $m^3$. Sensitivity has been performed to see the effect of tank size on the BOG generation in the storage tank. Figure 11 shows the BOG generation

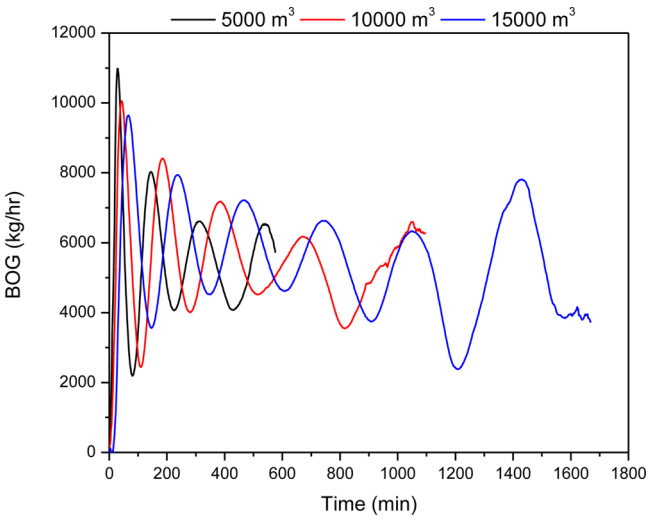



**Figure 11.** Effect of tank size on the BOG generation during loading mode.

during the tank loading mode of 5000, 10000, and 15000 $m^3$ storage tanks. It can be seen that in the case of the larger storage tank capacity, the BOG generation decreases for the same amount of input flow. In the case of the 10000 and 15000 $m^3$ tanks, the maximum BOG generated is 8.5% and 12.2% lower, respectively, compared to that produced during the loading of the 5000 $m^3$ tank. The reason is that as the tank filling starts, a larger vapor space is available that can adjust a certain amount of BOG in equilibrium with the liquid $CO_2$. However, as the tank continues to fill and the process tends to be more stabilized, the large capacity tank produces more BOG compared to the tanks with a smaller capacity because of the larger heat flux for the large capacity tank. Therefore, the recirculation flow rate of liquid $CO_2$ also increases for the large size tank. The recirculation flow rates after achieving steady state for 5000, 10000, and 15000 $m^3$ storage tanks are recorded as 113000, 115592, and 121910 kg/h, respectively.

**5.2. Effect of Storage Tank Operating Pressure.** The previous sections discussed in detail the relation between BOG generation and the tank pressure. This section will look at the effect of storage tank pressure on the BOG generation. A sensitivity analysis has been performed to analyze the effect of tank operating pressure on the generation of BOG. Figure 12

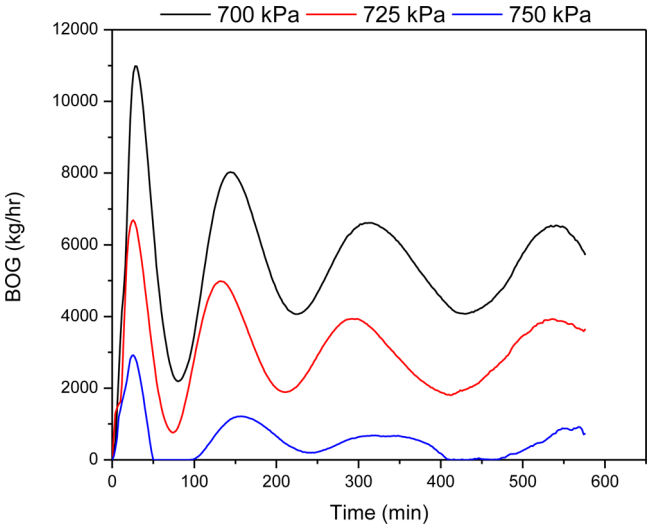



**Figure 12.** Effect of tank pressure on the BOG generation during loading mode.

shows the BOG generation under tank operating pressures of 700, 725, and 750 kPa. The results show that increasing the tank pressure decreases the BOG generation. However, increasing the tank pressure will lead to higher thickness for the tank material, hence causing an increase in the CAPEX. As shown in Table 6, an increase in tank operating pressure from 7 to 8 bar can increase the tank material cost by 5.9 million Euros.[10] Therefore, selection of tank pressure is a trade-off between the operating energy required to handle BOG and the capital cost of the tank.

**5.3. Miscellaneous Factors.** A sensitivity analysis is done for some of the other parameters that influence the amount of BOG generation in the $CO_2$ terminal. First, a sensitivity was performed to see the effect of recirculation flow rate on the BOG generation in the storage tank. In the base case, the recirculation flow rate was calculated so that there is no vapor in the recirculation line going back to the storage tank. The


DOI: 10.1021/ie503696x
Ind. Eng. Chem. Res. XXXX, XXX, XXX−XXX

recirculation flow rate can be increased to decrease the BOG generation in the tank. However, an increase in the recirculation flow rate increases the pumping energy requirement. Therefore, any increase in the recirculation flow rate above minimum recirculation flow rate is a trade-off between BOG handling cost and recirculation pumping energy cost.

The length of the pipeline from the liquefaction plant to the $CO_2$ terminal and the length of the forward and recirculation pipelines can affect the BOG generation within the $CO_2$ terminal, and hence, the corresponding costs can vary. In the base case, a pipeline length of 3 km is assumed from the liquefaction plant to the $CO_2$ terminal, and pipeline lengths of 1 km are assumed for both the forward and recirculation pipeline. However, if any of these pipeline lengths are increased, the BOG generation within the $CO_2$ terminal will increase, causing an increase in operational energy requirements. Since an increase in the pipeline length will increase the heat flux through the terminal, more BOG will be generated.

## CONCLUSION

In this study, a $CO_2$ terminal design and its operation philosophy has been presented. On the basis of an assumed terminal annual capacity of 5 Mton, a reasonable equipment sizing and its design has been done. The design is then tested using a dynamic simulation to track the operational performance of the terminal. Under normal operating conditions, three modes namely loading, holding, and unloading have been considered to define the operational characteristics of this pressurized transport chain. In an event of $CO_2$ leakage or abnormal operational parameters, a logical ESD system has been defined to safely shutdown the terminal operations. Whole terminal operation is kept within an operating pressure range of 6.5−7.5 bar with the use of PI controllers. The base case, the result shows that the maximum BOG rate is 11000 kg/h at the beginning of the tank filling operation for a capacity of 5000 $m^3$ tank. The terminal pipelines are kept under necessary cryogenic conditions by continuous recirculation of liquid $CO_2$ in order to avoid vapor flashing in the pipelines. The recirculation flow rate during the unloading mode is almost 2.4 times more than during the loading mode. Sensitivity analyses were done for various design parameters and conclude that the BOG rate is a trade-off between the capital cost to reduce BOG generation and the operating cost to handle the generated BOG.

## AUTHOR INFORMATION

### Corresponding Author

*Tel.: +82-02-880-1887. Fax: +82-02-880-1887. E-mail: chhan@snu.ac.kr.

### Notes

The authors declare no competing financial interest.

## ACKNOWLEDGMENTS

This research was supported by the second phase of the Brain Korea 21 Program in 2014, by Engineering Development Research Center (EDRC) funded by the Ministry of Trade, Industry & Energy (MOTIE), by the Energy Efficiency & Resources Core Technology Program of the Korea Institute of Energy Technology Evaluation and Planning (KETEP) granted financial resource from the Ministry of Trade, Industry & Energy, Republic of Korea (Grant 20132010201760).

## NOMENCLATURE

$D_{opt}$ = Pipeline diameter
$L$ = Inside radius
$P$ = Design pressure
$Q_m$ = Mass flow rate
$t$ = Minimum tank thickness
$\rho$ = $CO_2$ density
$\mu$ = $CO_2$ viscosity

### Abbreviations and Acronyms

ASME: American Society of Mechanical Engineers
ASTM: American Society for Testing and Materials
BOG: Boil-off gas
BOR: Boil-off rate
CAPEX: Capital expenditures
CCS: Carbon capture and storage
EOR: Enhanced oil recovery
ESD: Emergency shutdown
IPCC: Intergovernmental Panel on Climate Change
LNG: Liquified natural gas
OP: Opening percentage
OPEX: Operating expense
PV: Process variable
SRK: Soave−Redlich−Kwong
VLE: Vapor liquid equilibrium
ZEP: Zero emission platform

## REFERENCES

(1) Edenhofer, O., Pichs-Madruga, R., Sokona, Y.; Farahani, E.; Kadner, S.; Seyboth, K.; Adler, A.; Baum, I.; Brunner, S.; Eickemeier, P.; Kriemann, B.; Savolainen, J.; Schlömer, S., von Stechow, C.; Zwickel, T.; Minx, J. C. *IPCC, 2014: Climate Change 2014: Mitigation of Climate Change*; Cambridge University Press: Cambridge, U.K., 2014.

(2) International Energy Agency, *CCS 2014 What Lies in Store for CCS?*; IEA Publications: Paris, 2014.

(3) Zahid, U.; An, J.; Lee, U.; Choi, S. P.; Han, C. Techno-economic assessment of $CO_2$ liquefaction for ship transportation. *Greenhouse Gases: Sci. Technol.* **2014**, *4* (6), 734−749.

(4) Aspelund, A.; Jordal, K. Gas conditioning: The interface between $CO_2$ capture and transport. *Int. J. Greenhouse Gas Control* **2007**, *1* (3), 343−354.

(5) Lee, U.; Yang, S.; Jeong, Y. S.; Lim, Y.; Lee, C. S.; Han, C. Carbon dioxide liquefaction process for ship transportation. *Ind. Eng. Chem. Res.* **2012**, *51* (46), 15122−15131.

(6) Chiyoda Corporation. *Preliminary Feasibility Study on $CO_2$ Carrier for Ship-Based CCS*; Global CCS Institute: Docklands, VIC, Australia, 2011.

(7) Alexandrakis, M. I.; Smart, B. S. Marine Transportation for Carbon Capture and Sequestration(CCS), Ph.D. Dissertation, Massachusetts Institute of Technology, 2010.

(8) Kim, J. *Design and Feasibility Study of a Floating Barge for $CO_2$ Liquefaction and Storage*. Masters Thesis, Korea Advanced Institute of Science and Technology, 2011.

(9) Lee, U.; Lim, Y.; Lee, S.; Jung, J.; Han, C. $CO_2$ storage terminal for ship transportation. *Ind. Eng. Chem. Res.* **2011**, *51* (1), 389−397.

(10) Vermeulen, T. N. Knowledge Sharing Report: $CO_2$ Liquid Logistics Shipping Concept (LLSC). *Overall Supply Chain Optimization*; Global CCS Institute: Docklands, VIC, Australia. http://cdn.globalccsinstitute.com/sites/default/files/publications/19011/co2-liquid-logistics-shipping-concept-llsc-overall-supply-chain-optimization.pdf (accessed June 21, 2011).

(11) Li, H. Thermodynamic Properties of $CO_2$ Mixtures and Their Applications in Advanced Power Cycles with $CO_2$ Capture Processes, Ph.D. Thesis, Royal Institute of Technology, 2008.

(12) Ellenberger, J. P.; Chuse, R., Carson, B. E. *Pressure Vessels: The ASME Code Simplified*; McGraw-Hill Professional: New York, 2004.



DOI: 10.1021/ie503696x
*Ind. Eng. Chem. Res.* XXXX, XXX, XXX−XXX

(13) Barron, R. F. *Cryogenic Systems*, 2nd ed.; Clarendon Press: Oxford, 1985.
(14) *Nickel Alloy Steel Plates*; The International Nickel Company, Inc: New York, 1972.
(15) Zhang, Z. X.; Wang, G. X.; Massarotto, P.; Rudolph, V. Optimization of pipeline transport for $CO_2$ sequestration. *Energy Convers. Manage.* **2006**, *47*, 702−715.
(16) Peters, M. *Plant Design and Economics for Chemical Engineers*, 5th ed.; The McGraw-Hill Companies: New York, 2003.
(17) Beggs, D. H.; Brill, J. P. A study of two-phase flow in inclined pipes. *JPT, J. Pet. Technol.* **1973**, *25* (05), 607−617.
(18) Kumana, J. D.; Kothari, S. P. Predict Storage-Tank Heat Transfer Precisely. *Chemical Engineer* **1982**, *89* (6), 127−132.
(19) The Linde Group: Cryogenic Standard Tanks. http://www.linde-engineering.com/internet.global.lindeengineering.global/en/images/P_3_3_e_12_150dpi19_5774.pdf (Accessed March 20, 2014).
(20) Green, D. W. *Perry's Chemical Engineers' Handbook*; McGraw-Hill: New York, 2008.
(21) Decarre, S.; Berthiaud, J.; Butin, N.; Guillaume-Combecave, J. L. $CO_2$ maritime transportation. *Int. J. Greenhouse Gas Control* **2010**, *4* (5), 857−864.
(22) Dobrota, Đ.; Lalić, B.; Komar, I. *Problem of Boil-off in LNG Supply Chain. Transactions on Maritime Science* **2013**, *2* (02), 91−100.
(23) Yoo, B. Y.; Choi, D. K.; Kim, H. J.; Moon, Y. S.; Na, H. S.; Lee, S. G. Development of $CO_2$ terminal and $CO_2$ carrier for future commercialized CCS market. *Int. J. Greenhouse Gas Control* **2013**, *12*, 323−332.


DOI: 10.1021/ie503696x
*Ind. Eng. Chem. Res.* XXXX, XXX, XXX−XXX